Healthcare Appointment System — Suitable Process Model and Why

Oct 13, 2026 · @Hunny

1. Chosen Model

The best-suited process model is **Agile (Scrum)** with iterative, incremental delivery: 2-week sprints, each ending in a working, tested increment of the system. A short upfront phase fixes the architecture, database schema and roles; features are then delivered sprint by sprint.

2. Why Agile Fits This Project

- **Modular features:** the system splits cleanly into auth, doctor management, slots, booking, notifications and admin, so each can ship and be tested on its own.
- **Three user groups:** patients, doctors and admins can review each increment early, so usability problems surface before the whole system is built.
- **Evolving requirements:** reminders, rescheduling rules and dashboards usually change after users see them; short sprints absorb that change cheaply.
- **Early risk reduction:** the hardest technical problem, double-booking under concurrent load, is built and load-tested in an early sprint rather than at the end.
- **MERN fits iteration:** React components, REST endpoints and Mongoose schemas are easy to change and redeploy, and CI/CD supports frequent releases.
- **Small team:** daily stand-ups, sprint reviews and a shared backlog suit a team of 3 to 6 developers with little overhead.
- **Quality:** testing happens inside every sprint, so defects are found while the code is still fresh.

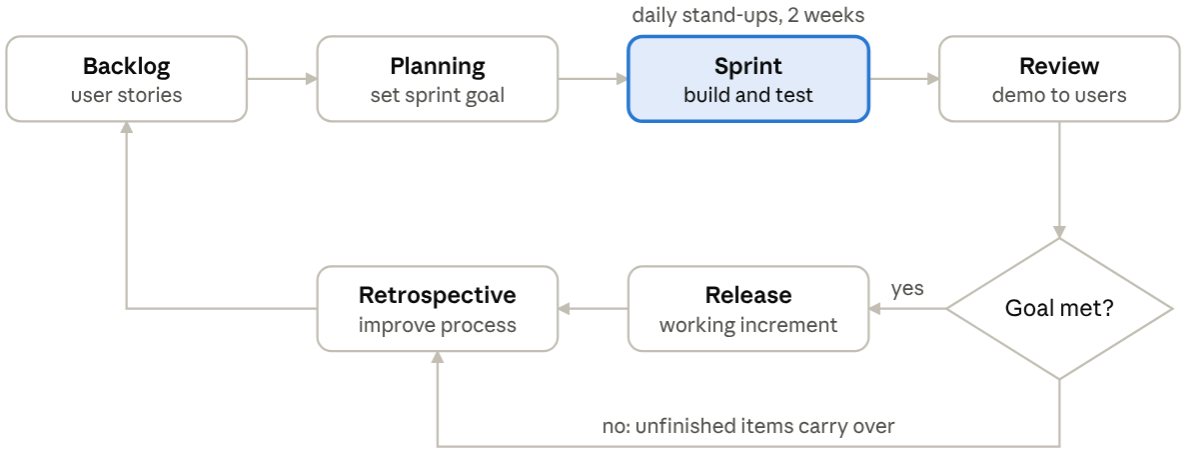
3. Alternatives Considered

Model	Strength	Why it is less suitable here
Waterfall	Simple, well documented	Needs all requirements frozen upfront; user feedback arrives only at the end
V-Model	Strong verification	Rigid and sequential; costly when requirements change

Model	Strength	Why it is less suitable here
Spiral	Strong risk analysis	Heavy process and cost for a web app with well-known technology
Prototyping	Quick user feedback	Lacks a delivery structure; risks throwaway or unfinished code
Agile (Scrum)	Frequent releases, feedback, adaptability	**Chosen:** best match for modular, evolving, multi-role software

4. Sprint Cycle and Delivery Plan

Every two-week sprint ends in a tested, releasable increment



Scrum cycle · 1 decision, 1 feedback loop

Each sprint takes backlog items to a reviewed increment; what is not finished returns to the backlog, and the retrospective feeds the next planning session.

Sprint	Focus	Output
0	Setup	Architecture, database schema, repository, CI pipeline
1	Authentication and roles	Registration, login, JWT, role-based access, user management
2	Doctors and slots	Profiles, specializations, availability, slot generation
3	Booking	Doctor search, booking, atomic conflict handling
4	Appointment lifecycle	Cancel, reschedule, status updates, history, dashboards

Sprint	Focus	Output
5	Notifications and admin	Reminders, job queue, admin dashboard, audit logs
6	Hardening and release	Load test at 1,000 users, security review, deployment